

ANI Enterprise ERP

Comprehensive Technical Documentation & Architecture Reference Guide

Secure Gateway & E2E Crypto

AES-256 encrypted payloads with HMAC-SHA256 request signatures and anti-replay timestamps routed through /v.

Full Trade & LC Lifecycle

End-to-end management of Proforma Invoices (PI), Packing Lists (PL), Trust Receipts (TR), Gate Passes, and Landed Cost (COGS).

Multi-Warehouse Inventory

Real-time FIFO & batch matching, border sales, damage isolation, inter-warehouse transfers, and live ledger reconciliation.

Client-Side Vector PDF Engine

Custom jsPDF generator with embedded Algerian, Cinzel, and Fraunces fonts for invoice and export documents.

Project: ANI Enterprise ERP

Stack: React 19 • Vite 7 • Express 5 • MongoDB 9 • Tailwind v4

Scope: Read-Only Technical Architecture

Version: 1.0.0 (Production Architecture)

1. Technology Stack & Architectural Overview

ANI Enterprise ERP is a full-stack, enterprise-grade management platform built specifically for international trade, import/export logistics, multi-warehouse stock reconciliation, letter of credit (LC) operations, and enterprise financial accounting.

Layer / Subsystem	Technology & Version	Technique / Exact Purpose in Project
Frontend Framework	React 19.2.0	Functional React components with hooks (useState, useMemo, useEffect, useRef) for single-page dynamic view switching.
Bundler & Dev Server	Vite 7.2.4	Lightning-fast HMR, Rollup production asset hashing (s/[hash].js), and development reverse-proxying.
Styling Architecture	Tailwind CSS v4 + Vanilla CSS	@tailwindcss/vite combined with custom component stylesheets for date pickers, stock modals, and LC management views.
Legacy Device Support	@vitejs/plugin-legacy	Legacy chunk generation and modern polyfills targeting Safari 7, iOS 7, Chrome 30, and legacy enterprise terminals (e.g. BlackBerry 10).
End-to-End Encryption	CryptoJS 4.2.0	AES payload encryption, HMAC-SHA256 digital signatures, and SHA-256 credential hashing on both client and server.
Data Communication	Axios + Fetch Shadowing	Custom Axios interceptors and global window.fetch wrapper redirecting all API queries through the secure /v gateway.
PDF Generation Engine	jsPDF 4.2.1 + AutoTable	Client-side vector PDF generation for invoices, gate passes, vouchers, PI/PL sheets with embedded base64 custom typography.
Backend Server	Node.js + Express 5.2.1	Async HTTP pipeline, internal gateway dispatcher, REST controllers, and 50MB payload parsing for bulk uploads.
Session & Authentication	express-session + MongoStore	Persistent session storage in MongoDB sessions collection, httpOnly secure cookies, and RBAC middleware.
Database & ODM	MongoDB + Mongoose 9.9.2	32 domain-driven collections, automated admin user seeding, and transactional updates.
Containerization	Docker Compose	3-tier isolation (erp_mongo, erp_server, erp_client) for local development and production deployments.

2. Security Architecture & The Encrypted Gateway (/v)

Enterprise Data Confidentiality

All business transactions, stock valuations, supplier pricing, and profit/loss calculations are shielded using an End-to-End Encrypted Gateway. Direct API paths (e.g. /api/sales) are never transmitted in cleartext over the network.

Gateway Communication Pipeline

- 1 Client-Side Interception (client/src/utils/api.js):**
When any component makes an `axios` or `fetch()` call to `/api/*`, the interceptor captures the target endpoint (`p`), HTTP method (`m`), and payload data (`d`).
- 2 AES Encryption & HMAC Signature:**
The package `{ p, m, d }` is encrypted via AES-256. A timestamp header (`X-Timestamp`) and a cryptographic HMAC-SHA256 signature (`X-Signature`) are generated and attached.
- 3 Gateway Dispatching (server/src/index.js & middleware):**
The client posts to `POST /v`. `securityMiddleware.js` validates the signature and timestamp (5-minute anti-replay protection), decrypts the payload, and rewrites `req.url`, `req.method`, and `req.body` for internal Express routing.
- 4 Encrypted Response Interception:**
Express route handlers execute standard `res.json()` calls. The middleware intercepts the output, encrypts the response into `{ payload: "", timestamp }`, and returns it. The client interceptor automatically decrypts it for the UI.

```
// Gateway transformation format in client/src/utils/api.js
const gatewayPayload = { p: originalUrl, // e.g. "/api/sales/create"
  m: originalMethod, // e.g. "POST"
  d: originalData // e.g. { customer: "XYZ", items: [...], total: 54000 } };
config.url = '/v';
config.method = 'post';
config.headers['X-Timestamp'] = timestamp;
config.headers['X-Signature'] = generateSignature(gatewayPayload, timestamp);
config.data = { payload: encryptData(gatewayPayload) };

```

3. Business Domain Modules & Technical Implementations

Letter of Credit (LC) & International Trade

Components: LCManagement, LCGatePass, LCExpense, MarginReturn, PI, PackingList, TRSetup, IPManagement

Technique: Complete LC opening lifecycle tracking, Margin return accounting, Port/C&F payment distribution, and Trust Receipt (TR) bank loan amortization.

Inventory & Multi-Warehouse Tracking

Components: StockManagement, WarehouseManagement, DamageManagement, TransferManagement, StockHistoryModal

Technique: Real-time FIFO/batch valuation in stockHelpers.js. Tracks physical stock, quarantine damage, branch transfers, and SKU transaction ledgers.

Sales, Border Sales & Commercial Billing

Components: SaleManagement, SalesReport, ProfitLoss, OrderManagement, ReturnProduct

Technique: Dynamic batch allocation, live invoice discounting, border sales reconciliation, customer returns (RMA), and real-time gross/net profit analysis.

Financials, Banking & Disbursements

Components: PaymentCollection, PayToCustomer, Bank, CnFPayment, InsurancePayment, CostOfGoods

Technique: Cheque clearing & bank ledger reconciliation, money receipt issuance, C&F agency duty settlement, and landed Cost of Goods Sold (COGS) analytics.

HR, System Access & Dynamic RBAC

Components: EmployeeManagement, SystemAccess, RoleCreation, Profile

Technique: Granular matrix-level access controls (View, Add, Edit, Delete, Special, ShowRate) across all 30 sub-modules managed via permissionHelper.js.

Backup, Restore & Notification Engine

Components: BackupRestore, NotificationMenu

Technique: IndexedDB File System Access integration for local snapshots, automated cron-like database dumps, and threshold alerts for low stock and LC expiry.

4. Backend Mongoose Schema Registry (32 Models)

Category	Model Files (server/src/models/)	Primary Business Responsibility
Core & System	User.js, Employee.js, Metadata.js, IpRecord.js, BackupSetting.js, Notification.js	Authentication credentials, employee profiles, IP access logs, dynamic roles, automated backup preferences, and system notifications.
Import & Trade Masters	Importer.js, Exporter.js, Supplier.js, Port.js, PI.js, PackingList.js, TRSetup.js	International buyers/sellers, ports of entry, proforma invoices, shipping container weight records, and trust receipt loan profiles.
LC Lifecycle	LCManagement.js, LCGatePass.js, LCExpense.js, MarginReturn.js, CostOfGoods.js	LC values, bank margins, custom clearance gate passes, auxiliary expense vouchers, and itemized landed costing.
Third-Party Agencies	CnF.js, CnFPayment.js, Insurance.js, InsurancePayment.js	Clearing & forwarding agency bills, marine insurance cover notes, and payment disbursement vouchers.
Inventory & Goods	Product.js, Warehouse.js, Stock.js, Purchase.js, PurchaseReceive.js, Damage.js, Return.js	SKU catalog, multi-warehouse stock allocations, purchase orders, goods receipt notes (GRN), damage quarantine, and returns.
Sales & Finance	Customer.js, Sale.js, Bank.js	Customer CRM ledgers, invoice lines, border sale logs, and bank account debit/credit transaction records.

5. Client PDF Generation Engines & Build Configuration

Client-Side Vector PDF Engines

All business documents are rendered natively in the client browser with zero server-side rendering latency using **jsPDF** and **jspdf-autotable**:

- **pdfGenerator.js (Master Document Engine):** Generates commercial sales invoices, money receipts, gate passes, delivery challans, profit/loss financial statements, and stock ledger sheets.
- **pipdfgenerator.js & pi2pdfgenerator.js:** Handles multi-template Proforma Invoices with automated currency conversion, HS Codes, and terms of delivery.
- **plpdfgenerator.js & pl2pdfgenerator.js:** Produces customs-ready Packing Lists with gross/net metric weight matrices and container numbers.
- **Embedded Typography:** Base64 encoded binary fonts (*Algerian, Cinzel, Fraunces*) loaded into jsPDF virtual filesystem for high-end typography.

Vite Production Build & Dev Setup

```
// Root script to spin up the entire ERP ecosystem: npm run dev // -> docker-compose up -d mongo
// -> concurrently "npm run dev --prefix client" "npm run dev --prefix server"
```

Configuration	Technical Specification
Client Port / Reverse Proxy	Vite running on port 3000 with proxy rules forwarding /api and /v to Express backend on port 5000.
Asset Naming Strategy	Rollup outputs bundles to s/[hash].js and s/[hash].css for URL obfuscation and cache busting.
Session Storage Engine	MongoStore connected to mongodb://mongo:27017/erp_db with session-only cookies (sameSite: 'lax').